You should Plugin your Telemed Ultrasound

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Introduction. Ultrasound has been used for speech studies for decades (Shawker & Sonies, 1985; Stone, 2005) In recent years Telemed micro ultrasound devices have become something of a norm **[cite 3-4 studies]**. While the manual says to use mains power (connect the device's power source to a wall socket) it does not say why, and the device will work with just power from the USB-connection to a computer. We discovered by accident why this recommendation exists: When running on USB power the ultrasound device will raise imaging power by steps during each recording. This is done to avoid setting off a short circuit panic mode in the USB port. Unfortunately, the stepping of intensity has consequences for data quality. This presentation provides a description of the consequences, a description of how to detect if old data has been compromised and a recommendation for future data recording.

Method.

- PD (Palo, 2019)
- DeepLabCut (Wrench & Balch-Tomes, 2022)
- slurpy

Results.

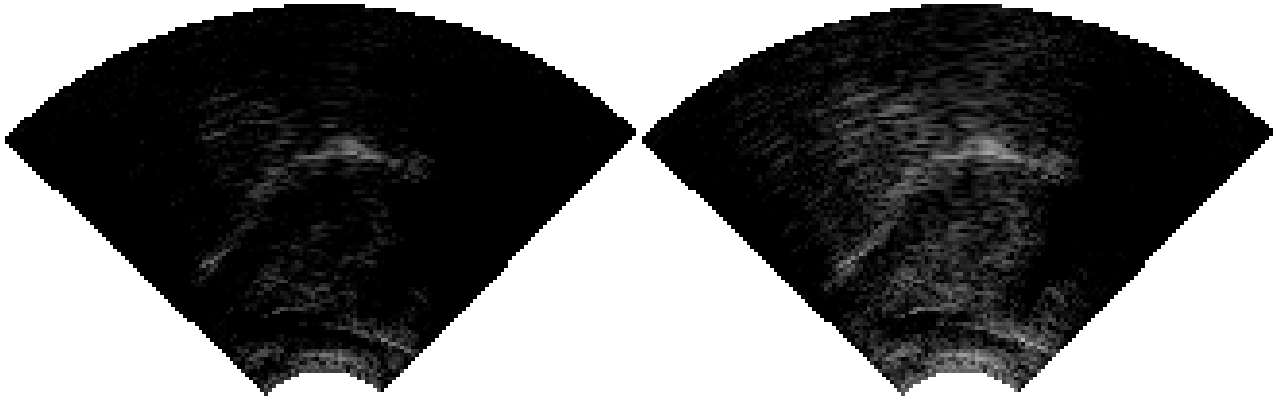


Figure 1: foo.

- example of bad data in PD
- example of sneakily bad data in PD
- similar for splining/image quality

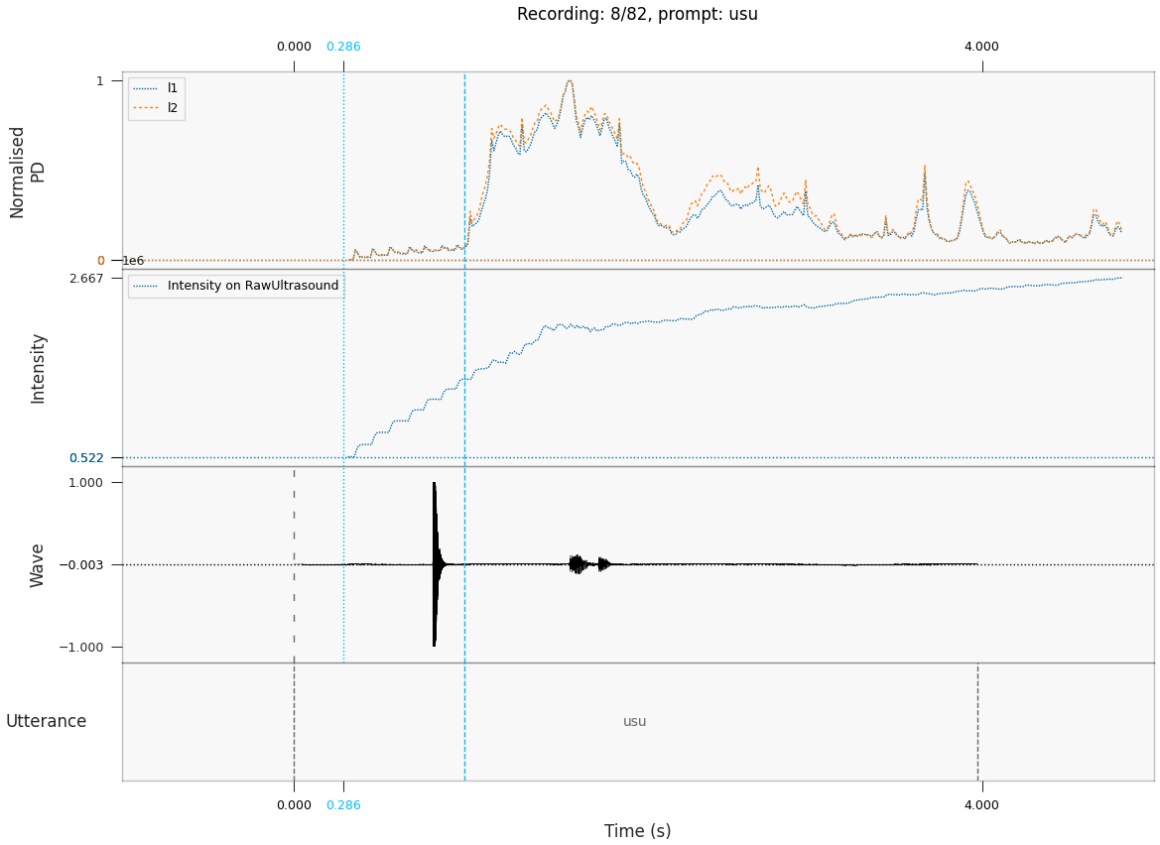


Figure 2: foo.

Conclusion. For new data the recommendation is simple: Plug the ultrasound machine into mains power when ever possible and if you need to run it with only USB power, make a note of that in your recording metadata / laboratory journal.

For old data, if you do not know that the device was plugged in to mains electricity,

References.

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